Proxy Courses

A peer-tutoring and advising platform for IUB students

Progress review · Tasks 1–3: Software Process, Requirements, Design

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CONTEXT

The problem and our answer

- Problem
  - Learners struggle in specific IUB courses; coaching is costly and off-syllabus
  - Seniors who aced the course are found only by luck (Facebook groups)
  - No trust: no grade check, no reviews, no refunds
  - Faculty and club advice is scattered

- Proxy Courses
  - Tutors with an admin-verified A / A- in the course
  - ৳500 semester subscription; 70/30 split (80/20 for top tutors)
  - Free intro video + free live demo before paying; clear refund rule
  - Free booking with faculty advisors and club executives
  - Responsive web app first; one API ready for mobile later

STATUS

Where we are

Task	Marks	Status
1 Software Process	10	Done: 3 / 3 items, 1 diagram
2 Requirements Engineering	22	Done: 16 / 16 items, 6 diagrams
3 Software Design	16	Done: 9 / 9 items, 6 diagrams
4 Product Metrics	14	Next
5 Quality Attributes	10	Planned
6 Testing	18	Planned (test plan drafted)
7 Project Management	18	Planned
8 Risk Management	12	Planned (6 risks drafted)

- Also written
  - PRD: 36 FRs, 12 NFRs
  - Architecture, data model, API
  - Roadmap M1–M4
  - Stack ADR: Next.js + PostgreSQL
  - Payments ADR: mock gateway → SSLCommerz
- 48 of 120 marks' worth of work drafted

Task 1 · Choosing the process model

Criterion	Weight	Waterfall	Spiral	Scrum + XP
C1 Copes with changing requirements	0.25	1	4	5
C2 Early working software	0.20	1	3	5
C3 Fits 14 weeks, task every 2 weeks	0.20	3	2	5
C4 Produces graded documents	0.20	5	4	3
C5 Risk control, small team	0.15	2	5	4
Weighted score	1.00	2.35	3.55	4.45

- Chosen: Scrum + XP + MoSCoW
  - Pricing and refund rules will change after interviews
  - 7 two-week sprints = 7 task deadlines
  - XP: pair work, PR review, CI on every push
  - Weak spot (docs) fixed: each sprint's Definition of Done includes its task document

Task 1 · Tailored life cycle

Proxy Courses: tailored life cycle (Scrum + XP, 14 weeks, 7 sprints, 2 exam gates)

	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Sprint (time-box)	**S1 · 2 weeks** Inception & process		**S2 · 2 weeks** Requirements		**S3 · 2 weeks** Design & metrics		**S4 · 2 weeks** Quality + build 1		**S5 · 2 weeks** Testing + build 2		**S6 · 2 weeks** Project mgmt + build 3		**S7 · 2 weeks** Risk & release	
Lectures	L1–2		L3		L4		L5		L5		L6–7		L8	
Course task due	**Task 1 · Process**		**Task 2 · Requirements**		**Task 3 + Task 4**		**Task 5 · Quality**		**Task 6 · Testing**		**Task 7 · Estimation**		**Task 8 · Risk**	
Artefacts	Proposal Process-model choice Life-cycle plan		PRD v1.0 / SRS Use cases, ER, state, DFD		Module design Wireframes FP count		Quality scenarios Availability targets		Test plan Test cases (WB/BB)		COCOMO, PERT WBS, Gantt, EVA		Risk register, RMMM Final report	
Product build	Repo, CI skeleton		Clickable mock-up		Schema + auth		Verification, offers		M1 core flow demo		M2 features		Release candidate	
Reviews	Sprint review + retrospective Faculty viva		Sprint review + retrospective Faculty viva		Sprint review + retrospective Faculty viva		Sprint review + retrospective Faculty viva		Sprint review + retrospective Faculty viva		Sprint review + retrospective Faculty viva		Sprint review + retrospective Faculty viva	

Inside every sprint: Planning (day 1) → daily stand-up → pair programming + PR review (XP) → CI tests on every push → review + retro (last day)

Throughout: **Configuration management (Lecture 6), all semester: Git branches, PR review, tagged baselines, GitHub Actions CI**

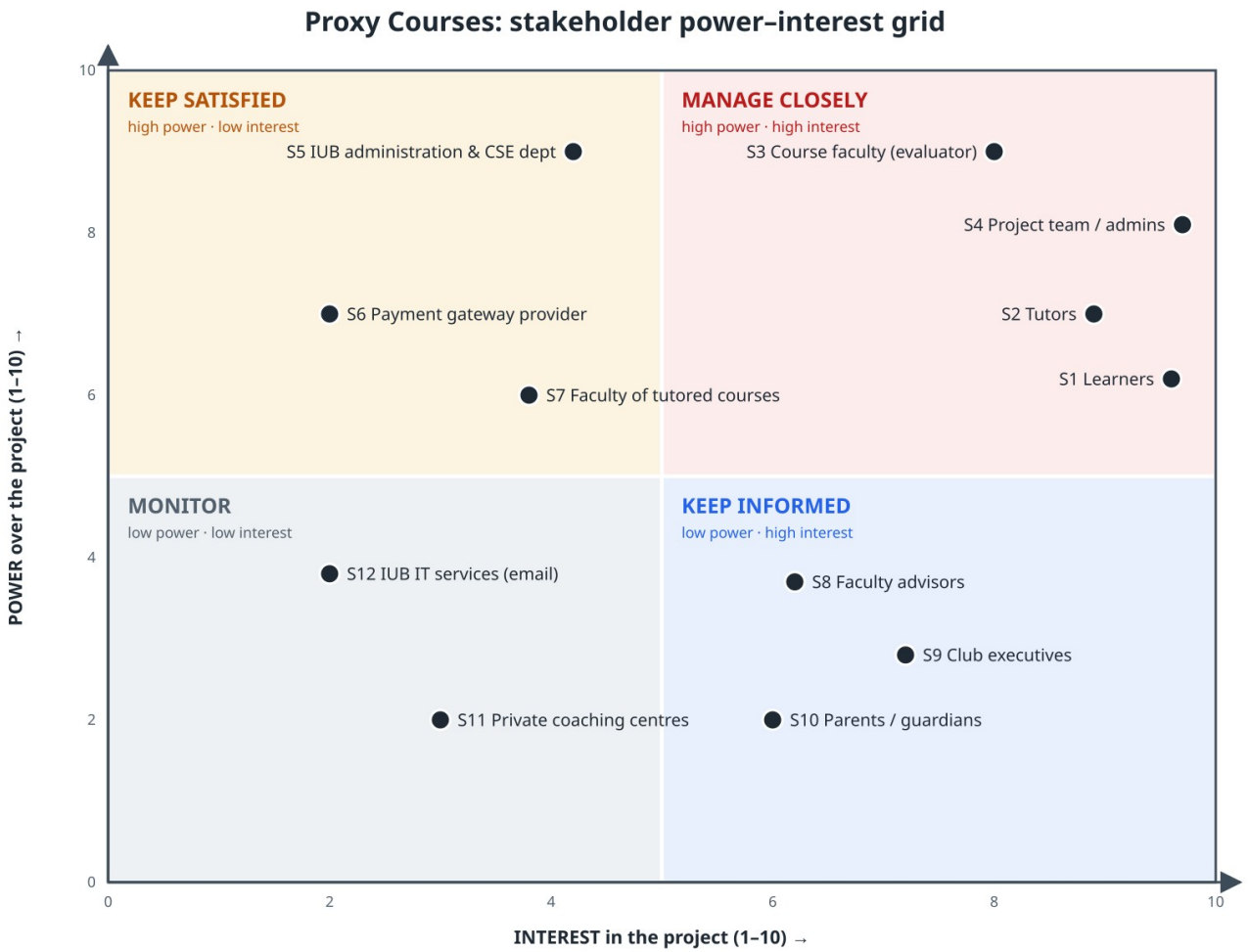
Milestones

- **M0 Proposal** — approved
- **M1 Requirements** — baseline v1.0
- **M2 Design** — baseline
- **G1 Midterm** — exam gate
- **M3 Core-flow** — prototype
- **M4 Test-ready** — release
- **M5 Final report** — + demo
- **G2 Final** — exam gate

◆ Milestone (artefact baselined in Git) ■ Exam gate (no sprint work due)

Week numbers are planned and move with the lecture schedule.

Task 2 · 12 stakeholders on a power–interest grid



- **Manage closely**: learners, tutors, course faculty, team
- **Keep satisfied**: IUB administration, payment gateway, faculty of tutored courses
- **Keep informed**: advisors, club executives, parents
- **Monitor**: coaching centres, IUB IT
- **Elicitation**: 9 interviews · survey (≥ 50) · analysis of 30 Facebook posts

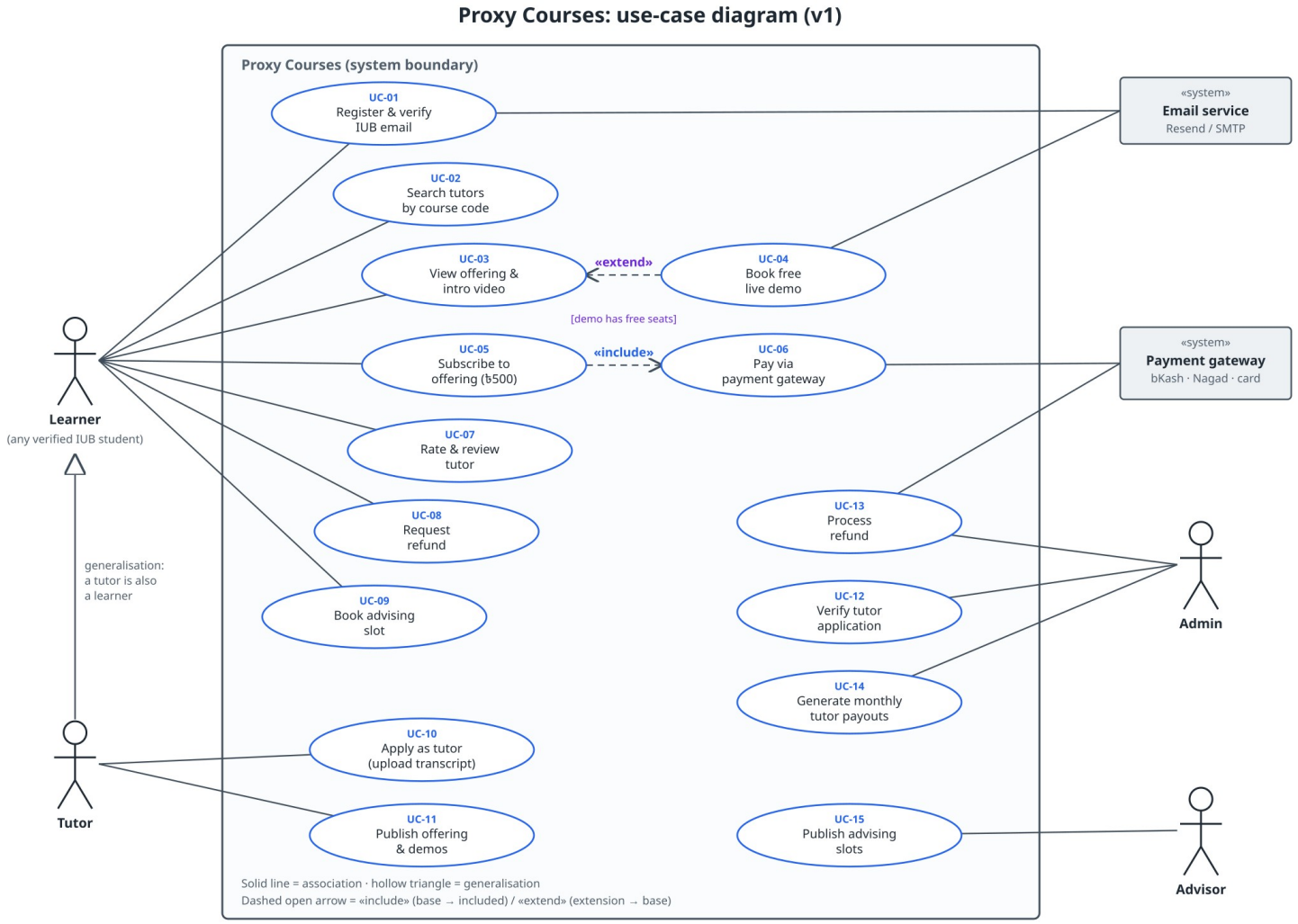
REQUIREMENTS

Task 2 · Requirements catalogue and user stories

Catalogue	Count
Functional requirements	36
Must / Should / Could	28 / 7 / 1
Won't (v1)	6 items
Non-functional, each with a metric	12
User stories (INVEST-checked)	6
Use cases	15

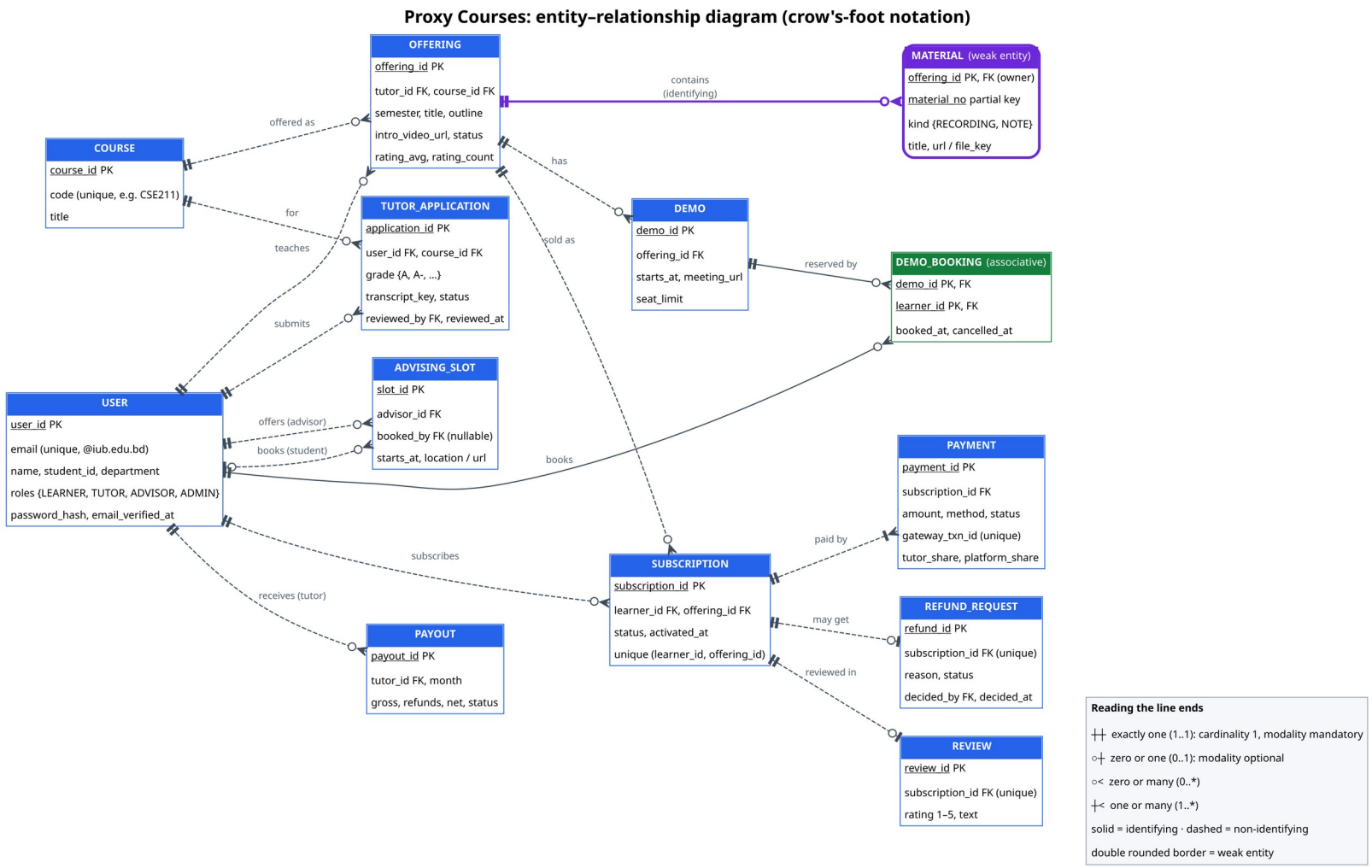
- Example NFR metrics
  - Search p95 ≤ 500 ms with 1,000 offerings
  - ≥ 99.0% monthly uptime (≤ 7.3 h down)
  - 0 transcripts reachable by other users
  - WCAG 2.1 AA: 0 serious axe-core issues
- INVEST caught 2 problems
  - US-3 too big → split into mock-pay and real gateway
  - US-5 depended on US-4 → seeded test data

Task 2 · Use-case diagram (15 use cases, 6 actors)



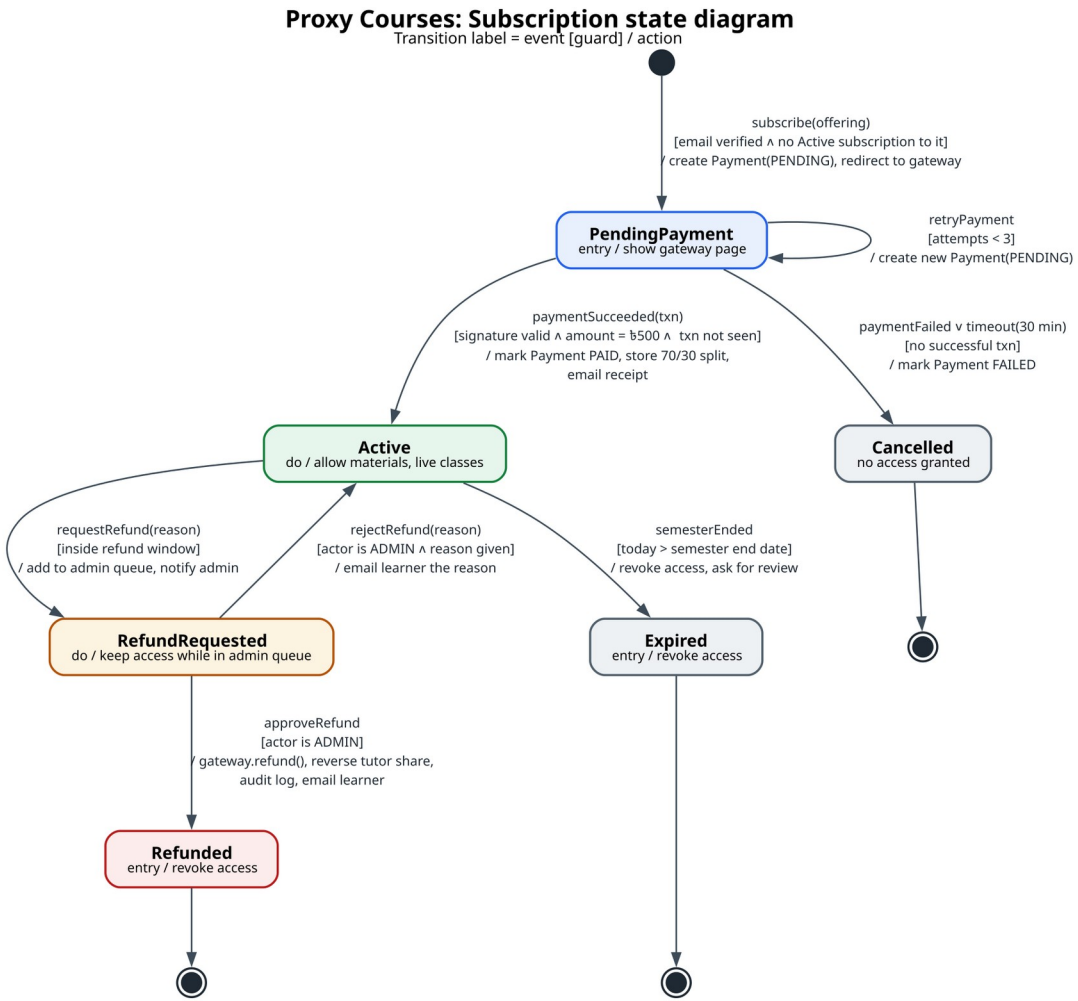
- **«include»**: Subscribe always includes Pay via gateway
- **«extend»**: Book free demo extends View offering [seats free]
- **Generalisation**: a tutor is also a learner
- **Fully dressed**: UC-05 Subscribe, with 6 extensions (failed payment, forged callback, duplicate callback…)

Task 2 · ER diagram (crow's foot)



- **13 entities**, PKs underlined
- **Weak entity**: MATERIAL (owner OFFERING + material_no)
- **Associative**: DEMO_BOOKING
- **Rules in the model**: REVIEW and REFUND hang off SUBSCRIPTION (0..1), so only subscribers can review

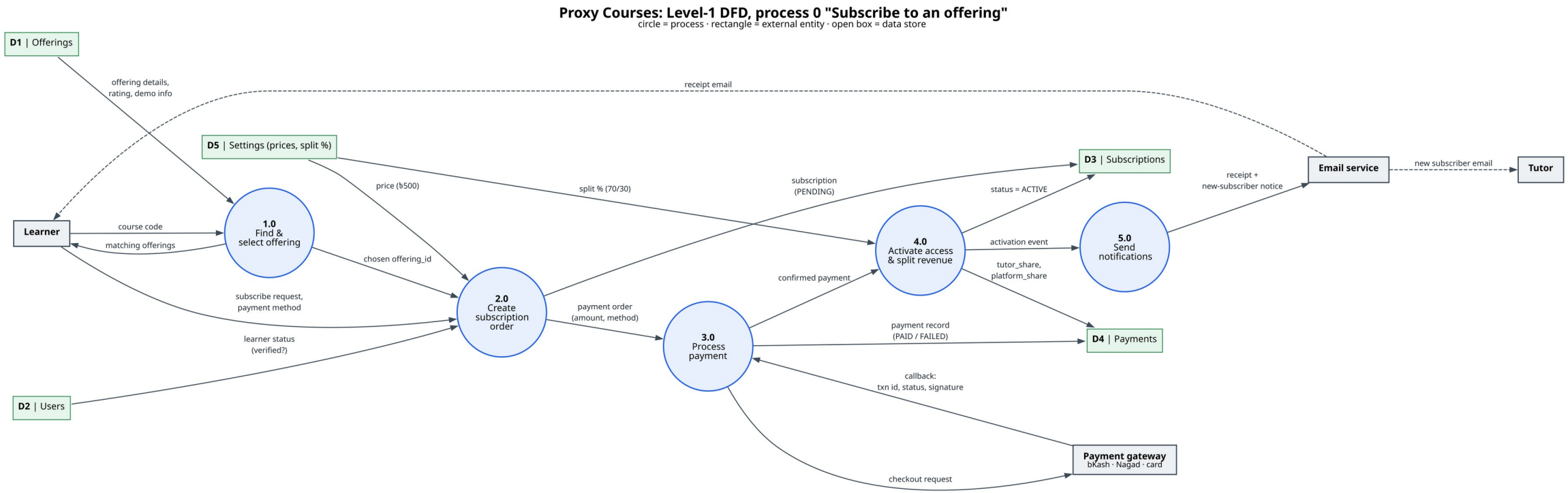
Task 2 · Subscription state diagram



- **6 states**: PendingPayment, Active, Cancelled, RefundRequested, Refunded, Expired
- Every transition = event [guard] / action
  - paymentSucceeded [signature valid ∧ amount = ৳500 ∧ txn new] / mark PAID, store split
  - requestRefund [inside window] / add to admin queue
- Guards prevent double charging and access without payment

REQUIREMENTS

Task 2 · Level-1 DFD: “Subscribe to an offering”



5 processes (find → order → pay → activate & split → notify) · 4 external entities · 5 data stores · flows balance with the context-level process

REQUIREMENTS

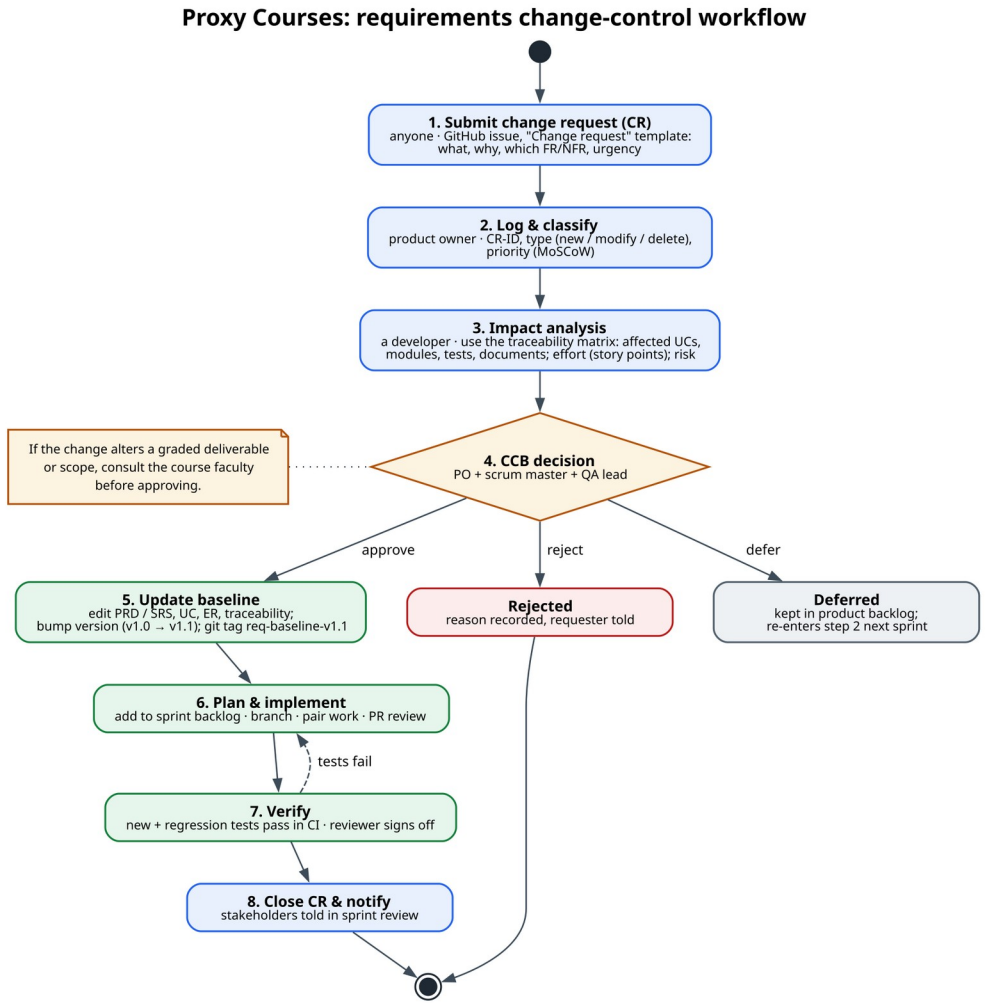
Task 2 · Negotiation, review and traceability

- Win-win 1: refunds vs tutor income
  - Full refund within 14 days and before the 3rd live class; later only by dispute (to confirm with interviews)
- Win-win 2: trust vs transcript privacy
  - Public sees “✓ Verified A in CSE211”; only admins see the transcript
- 8-question review on FR-16, FR-25, NFR-01
  - Found 3 ambiguities, 2 gaps, 1 conflict, all fixed

Req.	Use case	Module	Test
FR-16	UC-02	M4 Discovery	TC-06
FR-19	UC-04	M5 Booking	TC-07
FR-22	UC-06	M7 Payments	TC-09
FR-25	UC-08	M6 Subscriptions	TC-12
NFR-05	UC-10/12	M2, M12	TC-18

… 21 rows in total

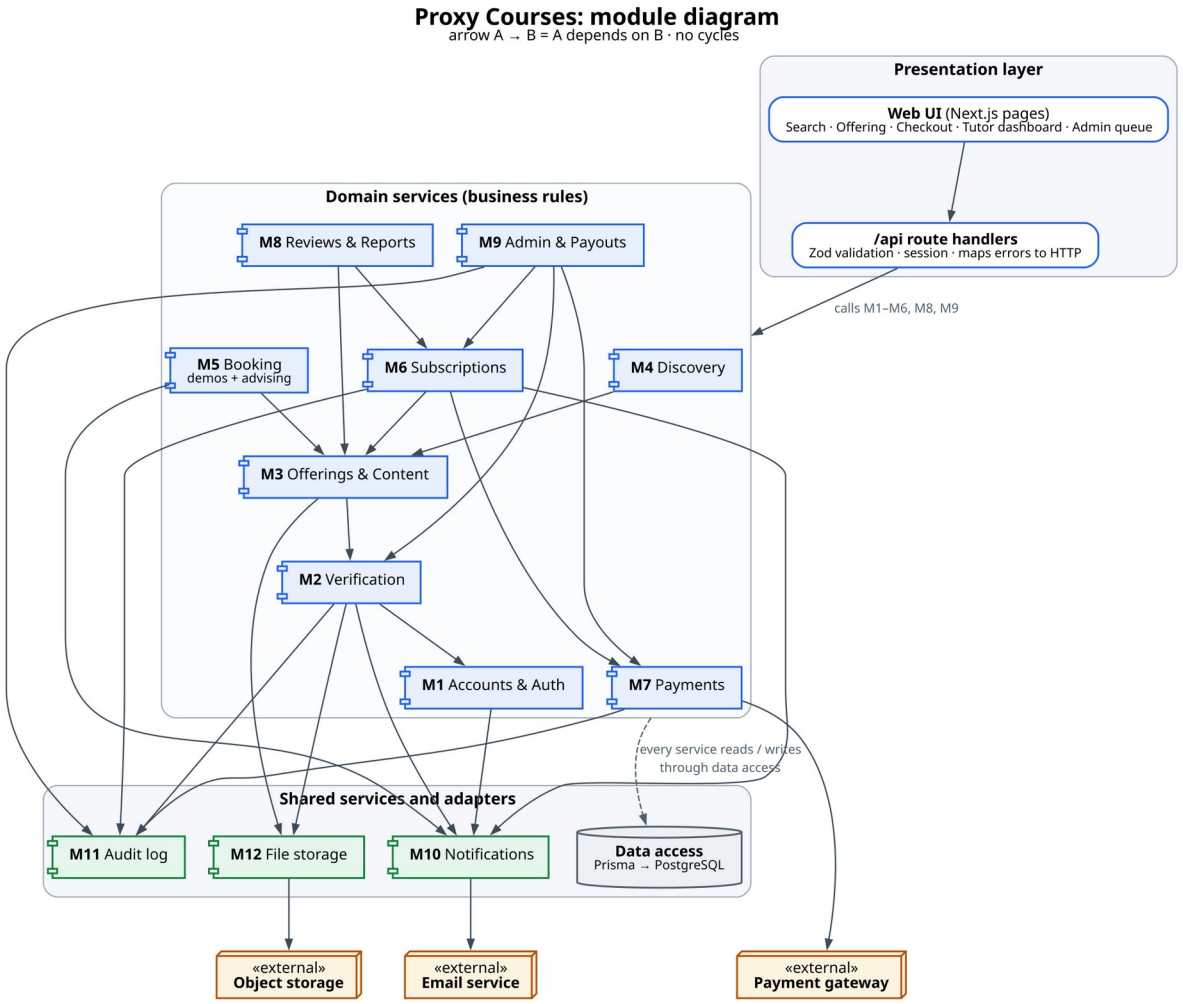
Task 2 · Change control and the requirements baseline



- **Baseline v1.0**: PRD + Task 2 models, approved at the Sprint 2 review
  - Frozen as git tag req-baseline-v1.0
- **After baselining**: change only through a CR
  - Impact analysis uses the traceability matrix
  - CCB: product owner + scrum master + QA lead
  - Approved → v1.1 with a new tag
- Scope or graded-deliverable changes are checked with the faculty first

DESIGN

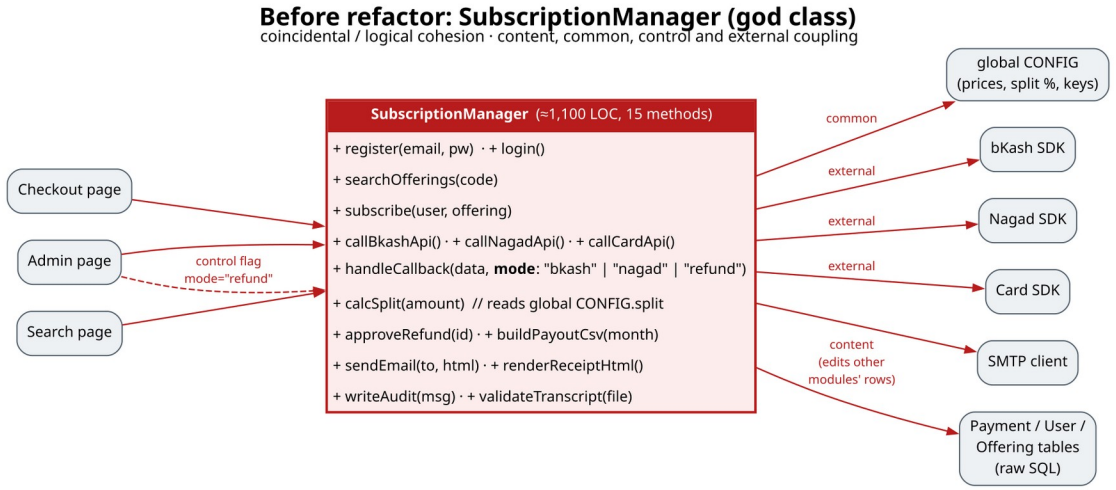
Task 3 · 12 modules in three layers

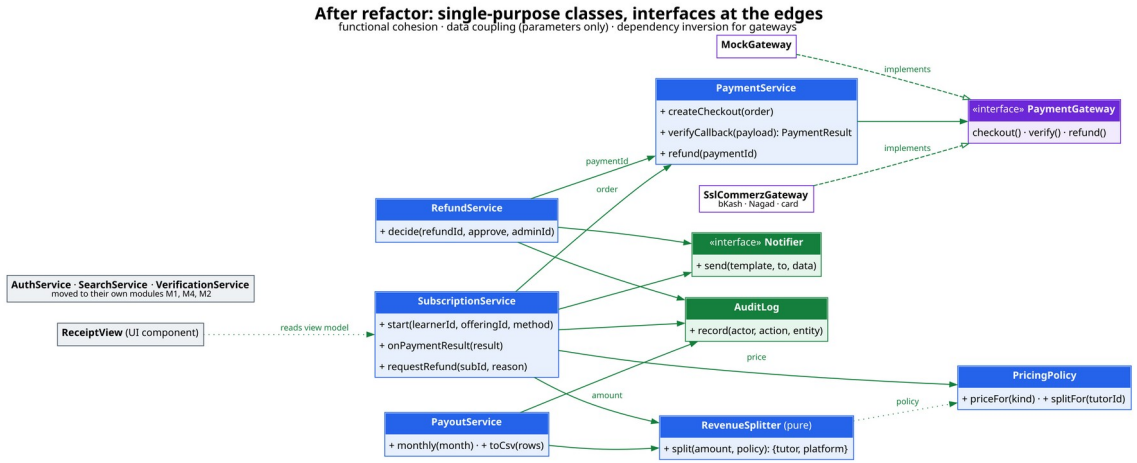


- **Each module**: responsibility, interface, hidden information, dependencies
- **No cycles**: M6 Subscriptions → M7 Payments, never back
- **Hidden**: only M7 knows which gateway is used
- **Cross-cutting**: roles, audit log, validation, error mapping

DESIGN

Task 3 · Fixing a god class





- **Before**: SubscriptionManager had coincidental and logical cohesion, and content, common, control and external coupling (15 methods, 9 collaborators)
- **After**: functional cohesion, data coupling and a PaymentGateway interface (mock or SSLCommerz). The control flag is replaced by polymorphism.

DESIGN

Task 3 · Mandel's golden rules, personas, task analysis

Golden rule	In Proxy Courses
User in control	× on every checkout step; cancel demo; withdraw refund
Reduce memory load	Step 2 of 3; refund as a date; last method pre-selected
Consistent interface	One blue bottom button; same status colours; same words

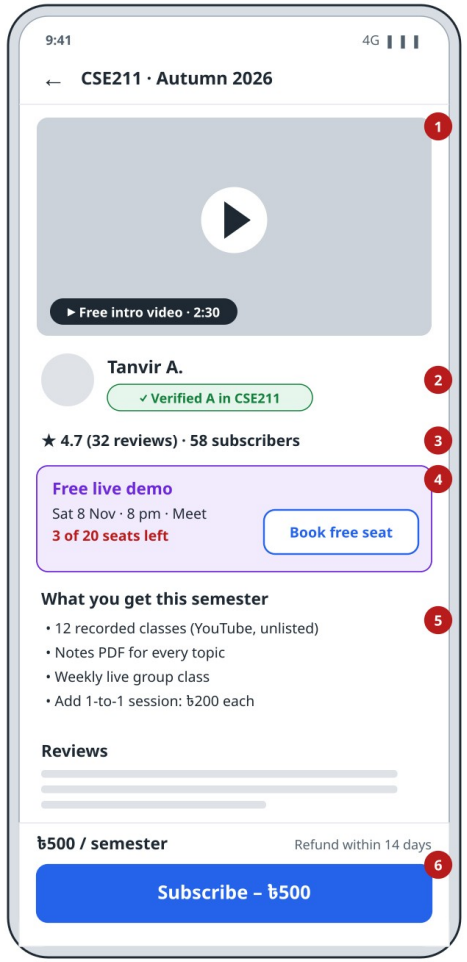
Persona	Goal
Nusrat · learner	Pass CSE211, spend ≤ ৳1,000
Tanvir · tutor	Earn about ৳10k a semester
Dr. Farhana Rahman · advisor	Stop the email ping-pong

Task analysis (HTA) of “Subscribe to help for CSE211”: 4 sub-goals, 11 leaf steps (12 with demo)
→ comparison data on result cards · one-screen checkout · demo email links straight to checkout

DESIGN

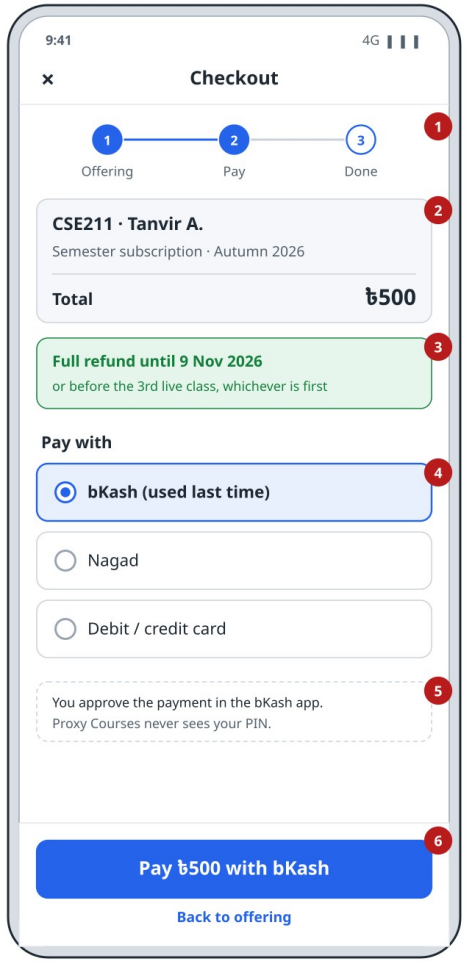
Task 3 · Annotated wireframes: offering page and checkout

Wireframe 1: Offering page (mobile, 390 × 844)



1. **Free intro video first**
   A free look before paying (proposal goal). Large 16:9 thumbnail with the length shown, so learners know the cost in time.
2. **Privacy-preserving trust badge**
   Shows only "Verified A in CSE211", never the transcript (win-win 2.13, NFR-05). Green = verified, used the same way everywhere.
3. **Social proof in one line**
   Rating, review count and subscribers. Recognition rather than recall: no need to open another page to compare tutors.
4. **«extend» Book free demo (UC-04)**
   Only shown when the demo has seats. Seats left in red creates honest urgency. Secondary (outline) style so it never competes with Subscribe.
5. **Plain list of what ৳500 buys**
   Sets expectations and cuts refund requests. The add-on price is shown up front, with no surprises at checkout.
6. **One primary action, in the thumb zone**
   Sticky full-width 358 × 52 px button at the bottom; the label states the price. See the Fitts's Law calculation in 3.9.

Wireframe 2: Checkout (mobile, 390 × 844)

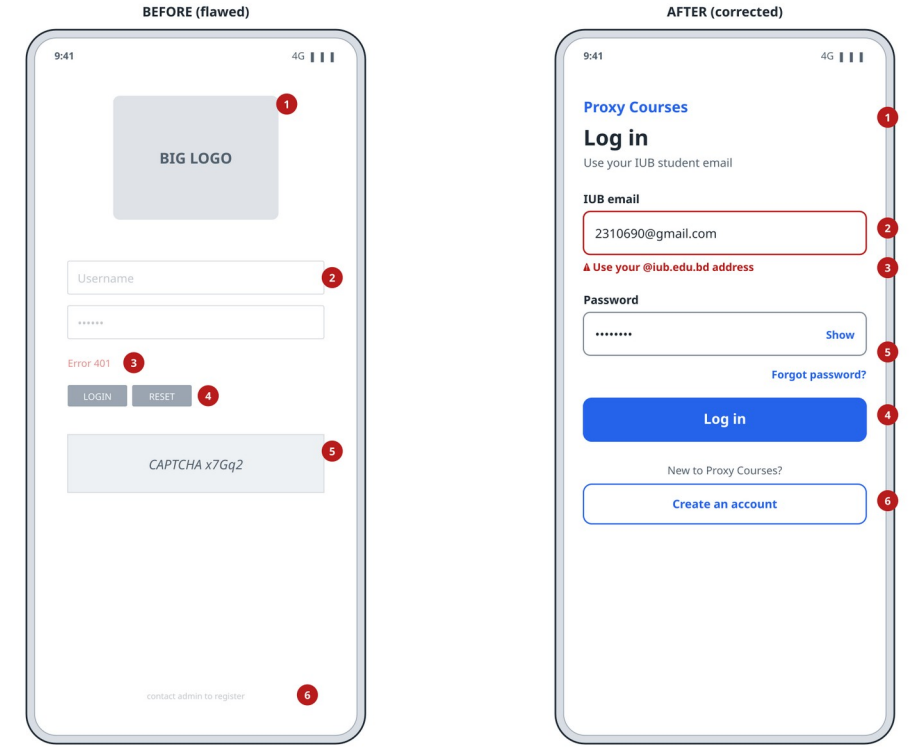


1. **Progress + exit always visible**
   "Step 2 of 3" lowers memory load; × lets the user leave at any time with nothing charged (Mandel rule 1: user in control).
2. **Summary in the user's words**
   Course code, tutor, semester and total, so the learner can confirm without going back to the offering page.
3. **Refund rule as a real date**
   Turns the rule (FR-25, win-win 2.13) into a date the learner doesn't have to work out. Green box = reassurance, same meaning across the app.
4. **Sensible default, easy to change**
   Last-used method is pre-selected; each option is a 52 px tall row, easy to tap with a thumb.
5. **Trust note**
   Says where approval happens and that card or PIN data is never stored (NFR-06).
6. **Button says exactly what happens**
   "Pay ৳500 with bKash" updates with the method chosen. Same place and style as Subscribe on the previous screen (Mandel rule 3: consistency).

DESIGN

Task 3 · Corrected login screen and Fitts's Law

Login screen: flawed version and corrected version



Problem (before)

1. Huge logo pushes the form down
2. Placeholder-only "Username": email or student ID?
3. "Error 401", pale red, far from the field
4. Tiny 72×26 LOGIN next to RESET (wipes the form)
5. CAPTCHA on every login; no show-password
6. No sign-up or reset path; low-contrast text

Fix (after)

Small wordmark; the form starts in the top third

Visible label "IUB email" + example; placeholder never replaces the label

Plain-language error under the field it belongs to, with the fix

One full-width 342×52 primary button; RESET removed

"Show" toggle; CAPTCHA only after 5 failed tries (rate limit)

"Forgot password?" + "Create an account"; all text ≥ 4.5:1 contrast

$MT = a + b \cdot \log_2(D/W + 1)$	A: small, top right	B: sticky bottom
Size	90 × 36 px	358 × 52 px
D (from (195, 500))	450.7 px	304.0 px
$ID = \log_2(D/W+1)$	3.76 bits	2.78 bits
MT (a = 0.10, b = 0.15)	0.66 s	0.52 s

- **B is 22% faster**, in the thumb zone and above the 44–48 px minimum
- **Login fixes**: real labels, an error under the field, one 342 × 52 button, no RESET, a sign-up path

PLAN

Next steps and questions for sir

- Next
  - Task 4: function points (feeds COCOMO in Task 7), OO metrics, Halstead
  - Run the interviews and survey → confirm the refund rule and price
  - Agree team roles: PO, scrum master, QA lead
  - Build M1: sign-up → search → demo → mock payment
- Questions for Masum sir
  - Are the swimlane, sequence and class diagrams required?
  - Is a full SRS (ISO 29148) required, or is the PRD enough?
  - Do we use UniRide's figures (activity list, Halstead) or our own?
  - Which week is each task due?